

Phase 3 Research Gap

Current zero-masking in transformed feature space is a weak tabular coalition approximation because it produces unrealistic standardized and one-hot feature patterns. On Covertypes, this matters because elevation and soil climate carry clear dependence structure.

The proposed solution is empirical-background masking: replace hidden original feature groups with values from real transformed training rows, then average coalition outputs across multiple sampled backgrounds. This yields a stronger data-aware value function for surrogate and InstaSHAP training.

This extension is intentionally scoped: it is not full conditional SHAP, but it directly addresses the zero-masking weakness and creates a fair before-vs-after experiment that can be reproduced from code.

Compact Before vs After

model	accuracy_mean	log_loss_mean	explanation_mae_mean	explanation_spearman_mean	coalition_mse_mean	explain_seconds_mean
instashap_zero	0.6098	0.9312	0.2805	0.4951	0.2863	0.0035
instashap_bg	0.6135	0.9356	0.3109	0.4518	0.3793	0.0050